Lab no 04: Implement 7-Segment Decoder on FPGA

The purpose of this Lab is to learn to: Implement the seven-segment decoder on FPGA. You will write the logic equation for each segment using Verilog bitwise operators.

Objective: a seven-segment run on FPGA (Here).

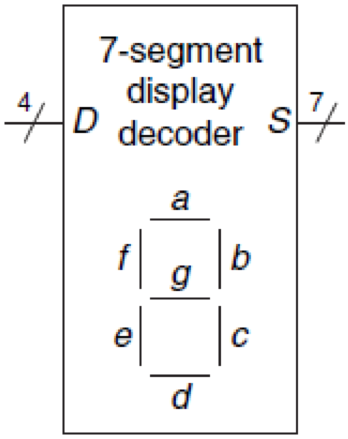
Refer to assignment 2, to review the logic equations of the seven-segment decoder.

Parts: -

Implement the seven-segment decoder using logic equations and run it on the FPGA.

Implement the seven-segment decoder using logic equations.

A seven-segment display decoder takes a 4-bit data input A, B, C, D and produces seven outputs to control light-emitting diodes to display a digit from 0 to 9. The seven outputs are often called segments a through g, as defined in Figure.



The truth table of the seven-segment decoder:

Digit	A	B	C	D	a	b	c	d	e	f	g
0	0	0	0	0	0	0	0	0	0	0	1
1	0	0	0	1	1	0	0	1	1	1	1
2	0	0	1	0	0	0	1	0	0	1	0
3	0	0	1	1	0	0	0	0	1	1	0
4	0	1	0	0	1	0	0	1	1	0	0
5	0	1	0	1	0	1	0	0	1	0	0
6	0	1	1	0	0	1	0	0	0	0	0
7	0	1	1	1	0	0	0	1	1	1	1
8	1	0	0	0	0	0	0	0	0	0	0
9	1	0	0	1	0	0	0	0	1	0	0

The logic equations of the seven-segment decoder

$$a = A + C + BD + \overline{B}\,\overline{D}$$

$$b = \overline{B} + \overline{C}\,\overline{D} + CD$$

$$c = B + \overline{C} + D$$

$$d = \overline{B}\,\overline{D} + C\,\overline{D} + B\,\overline{C}\,D + \overline{B}\,C + A$$

$$e = \overline{B}\,\overline{D} + C\,\overline{D}$$

$$f = A + \overline{C}\,\overline{D} + B\,\overline{C} + B\,\overline{D}$$

$$g = A + B\,\overline{C} + \overline{B}\,C + C\,\overline{D}$$

- **Bitwise operators**

&	Bitwise and / reduction and
|	Bitwise or / reduction or
^	Bitwise xor / reduction xor
~	Bitwise not

Verilog code for Decoder to 7 segments

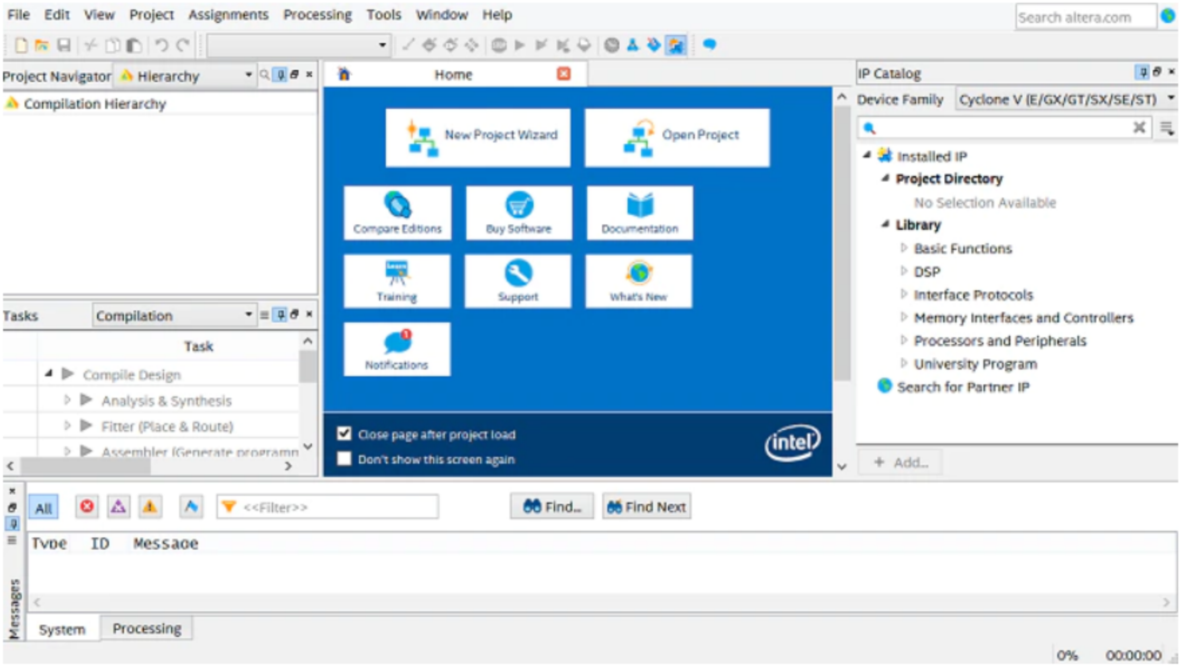
```
module decoder_7seg (A, B, C, D, led_a, led_b, led_c,
led_d, led_e, led_f, led_g);
      input A, B, C, D;
      output led_a, led_b, led_c, led_d, led_e, led_f,
             led_g;

      assign led_a = ~(A | C | B&D | ~B&~D);
      assign led_b = ~(~B | ~C&~D | C&D);
      assign led_c = ~(B | ~C | D);
      assign led_d = ~(~B&~D | C&~D | B&~C&D | ~B&C |A);
      assign led_e = ~(~B&~D | C&~D);
      assign led_f = ~(A | ~C&~D | B&~C | B&~D);
      assign led_g = ~(A | B&~C | ~B&C | C&~D);
endmodule
```

Run the seven-segment decoder on FPGA.

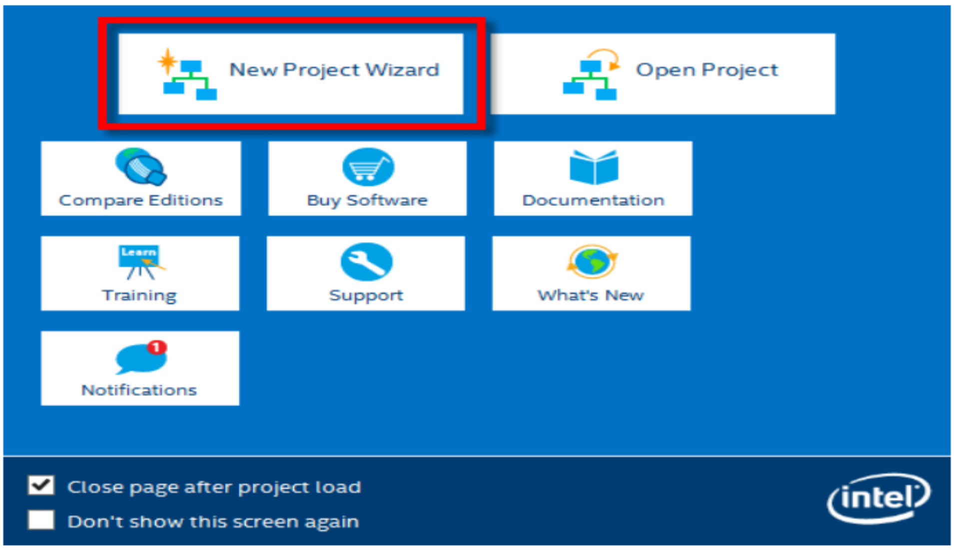
We will use a DE-10lite kit, Altera MAX 10 based FPGA board, check here. Check DE10-lite user manual (Here).
You will use **Quartus** to program the FPGA.

Quartus – Seven Segment Decoder Project Steps

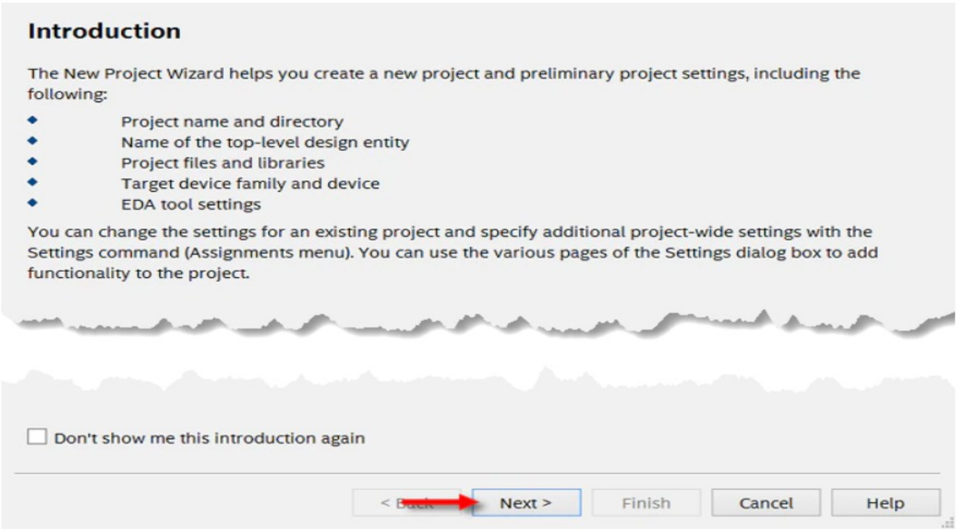
Step 1: Open Quartus.



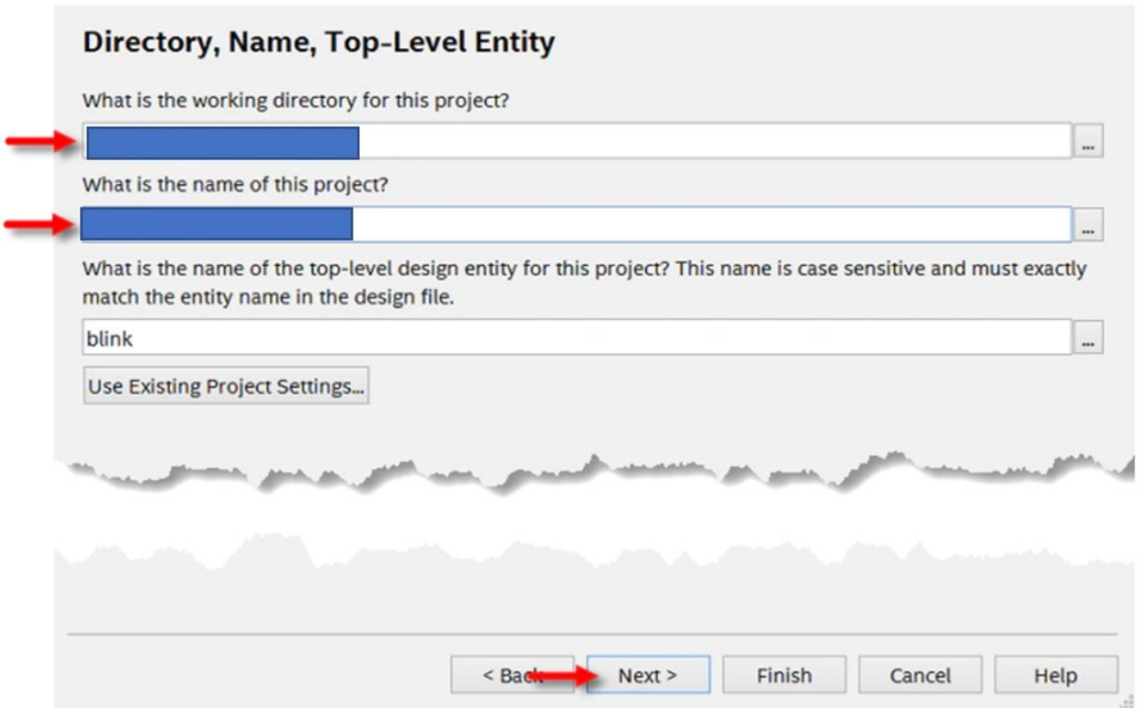
Step 2: Open a New Project Wizard.



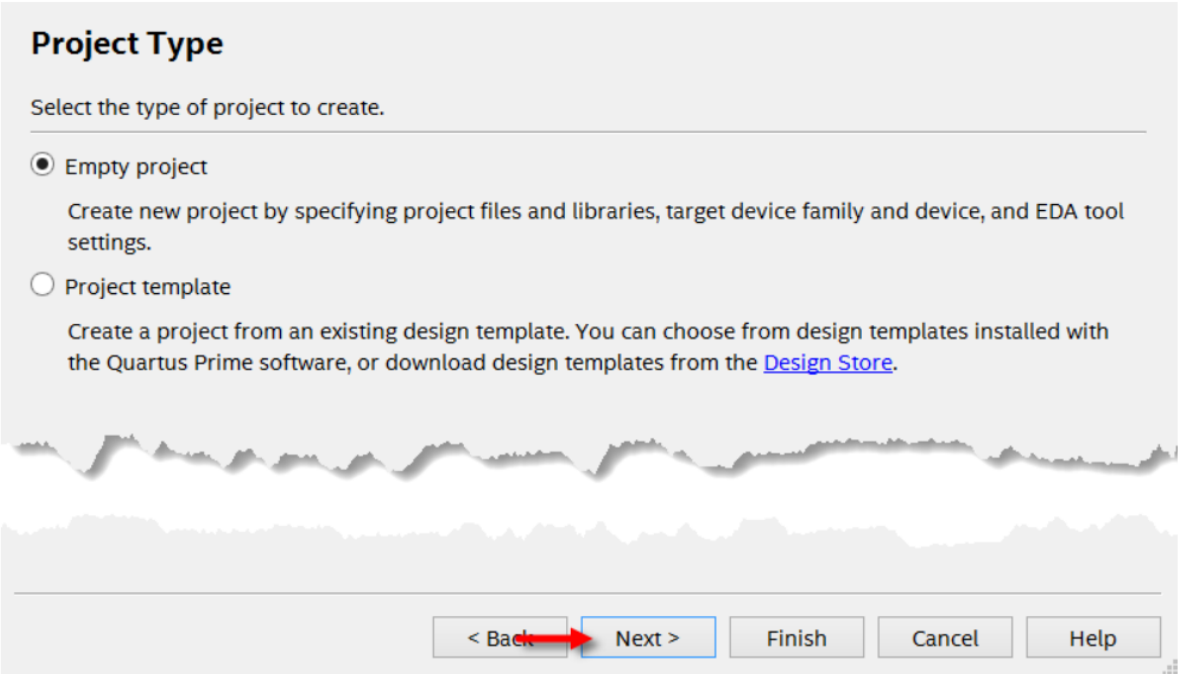
Step 3: Select Next



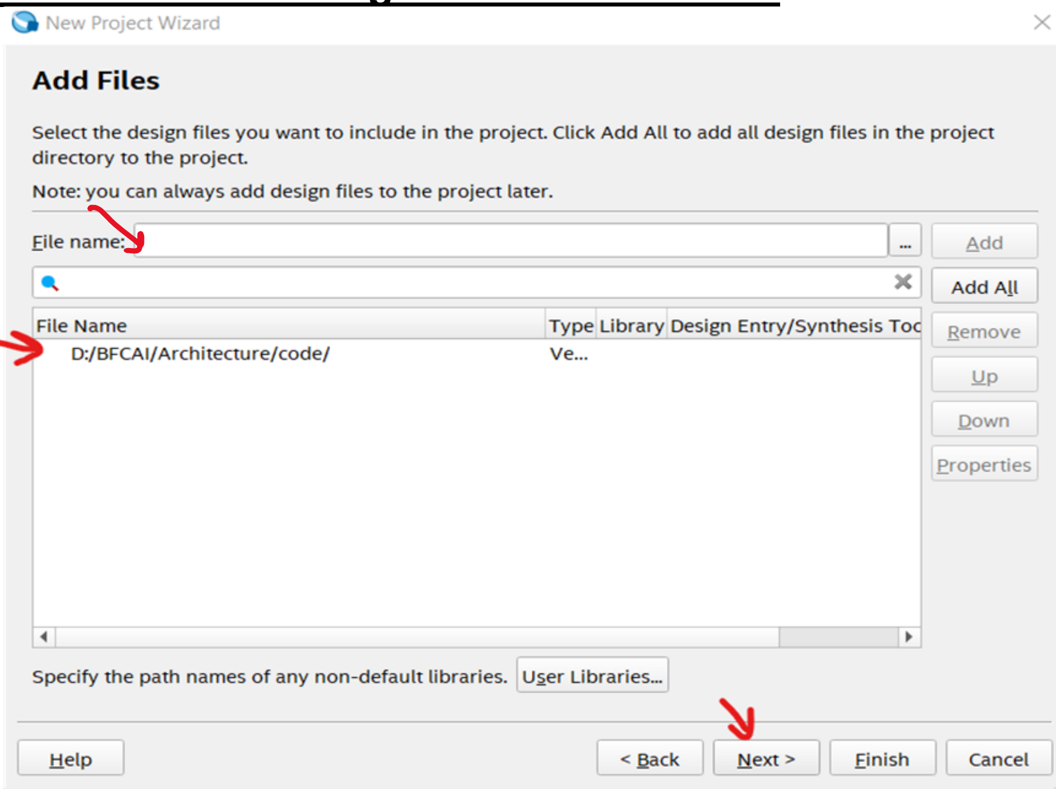
Step 4: Choose a directory to put your project under. you can place it wherever you want. Name the project as the name of the top-level module. You will name it **"decoder_7seg",** Select Next.



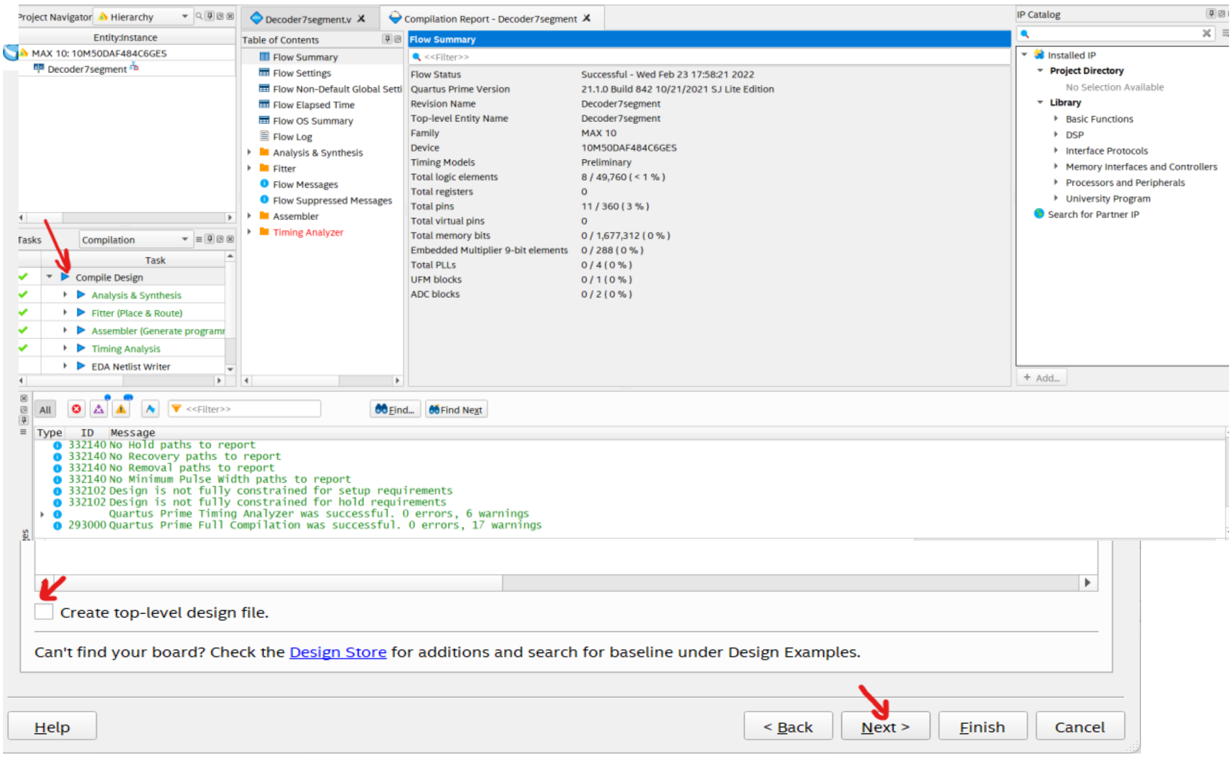
Step 5: Select Empty Project, and then click Next.



Step 6: If You want to add any files here. **Add the seven-segment decoder Verilog file And Click Next**.

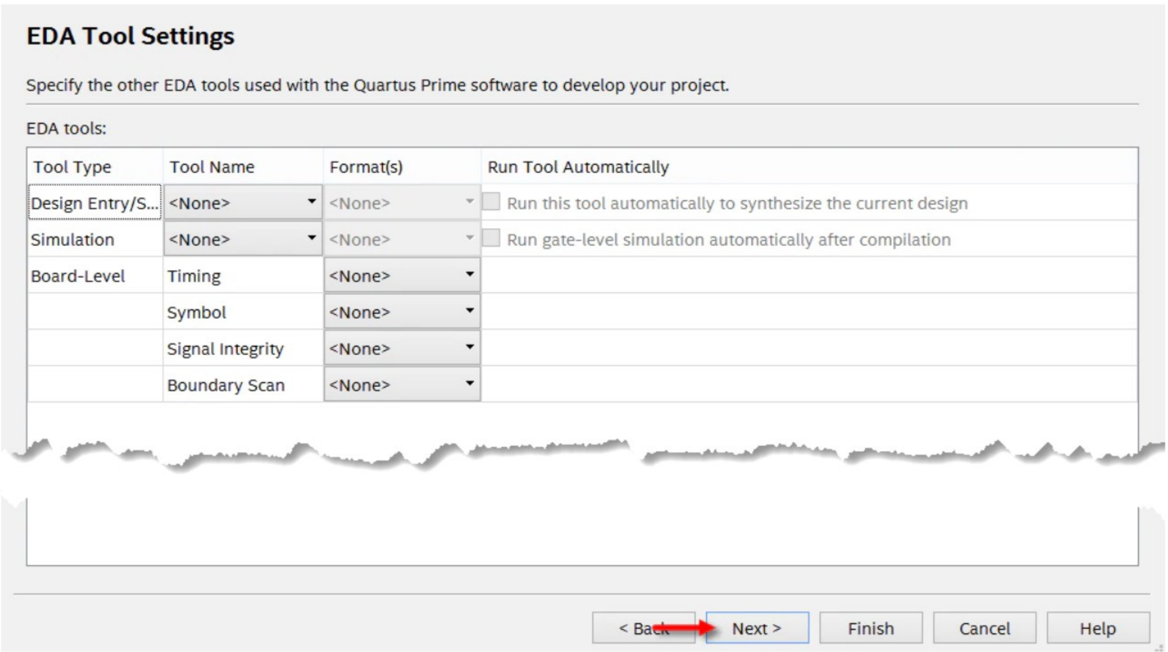


Step7: Board Settings, select our board **<u>DE10-lite</u>**, **<u>unmark</u>**



<u>“create top-level design file”</u>

<u>Step 8:</u> EDA Tool Settings then finish.



<u>Step 9:</u> compile the design

<u>Step 10:</u> Pin assignment on FPGA

In this step, you will assign inputs (A,B,C,D) to the switches,
And the outputs (led_a, led_b, led_c, led_d, led_e, led_f, led_g) to the first segment display.

To assign pins, refer to DE10-lite [FPGA user manual].

<u>User-Defined Slide Switch Section</u>

Figure 3-15 Connections between the slide switches and MAX 10 FPGA

Table 3-4 Pin Assignment of Slide Switches

Signal Name	FPGA Pin No.	Description	I/O Standard
SW0	PIN_C10	Slide Switch[0]	3.3-V LVTTL
SW1	PIN_C11	Slide Switch[1]	3.3-V LVTTL
SW2	PIN_D12	Slide Switch[2]	3.3-V LVTTL
SW3	PIN_C12	Slide Switch[3]	3.3-V LVTTL
SW4	PIN_A12	Slide Switch[4]	3.3-V LVTTL
SW5	PIN_B12	Slide Switch[5]	3.3-V LVTTL
SW6	PIN_A13	Slide Switch[6]	3.3-V LVTTL
SW7	PIN_A14	Slide Switch[7]	3.3-V LVTTL
SW8	PIN_B14	Slide Switch[8]	3.3-V LVTTL
SW9	PIN_F15	Slide Switch[9]	3.3-V LVTTL

<u>7-segment displays section</u>

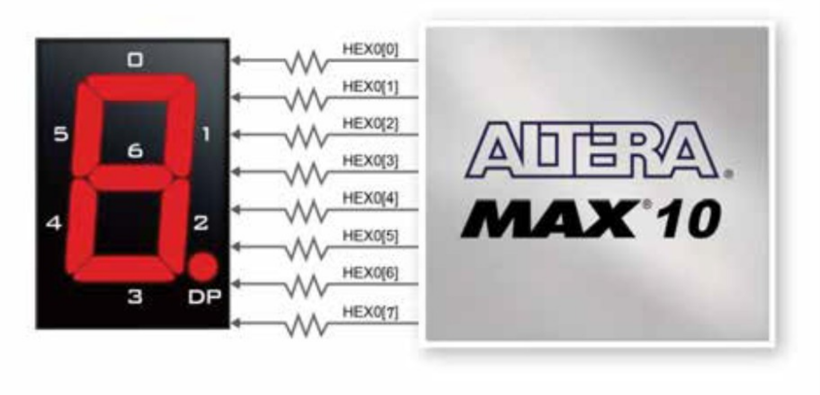
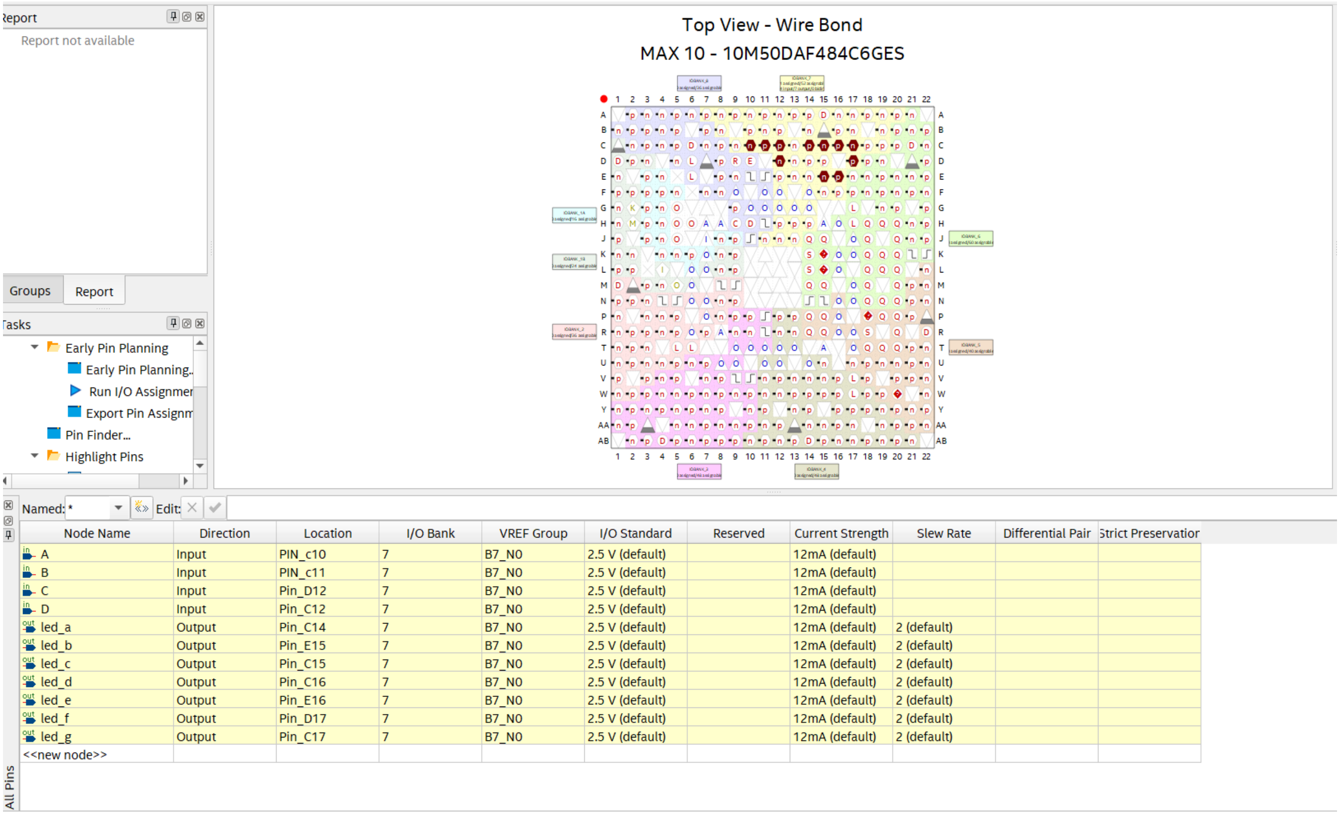


Figure 3-17 Connections between the 7-segment display HEX0 and the MAX 10 FPGA

Table 3-6 Pin Assignment of 7-segment Displays

Signal Name	FPGA Pin No.	Description	I/O Standard
HEX00	PIN_C14	Seven Segment Digit 0[0]	3.3-V LVTTL
HEX01	PIN_E15	Seven Segment Digit 0[1]	3.3-V LVTTL
HEX02	PIN_C15	Seven Segment Digit 0[2]	3.3-V LVTTL
HEX03	PIN_C16	Seven Segment Digit 0[3]	3.3-V LVTTL
HEX04	PIN_E16	Seven Segment Digit 0[4]	3.3-V LVTTL
HEX05	PIN_D17	Seven Segment Digit 0[5]	3.3-V LVTTL
HEX06	PIN_C17	Seven Segment Digit 0[6]	3.3-V LVTTL
HEX07	PIN_D15	Seven Segment Digit 0[7], DP	3.3-V LVTTL

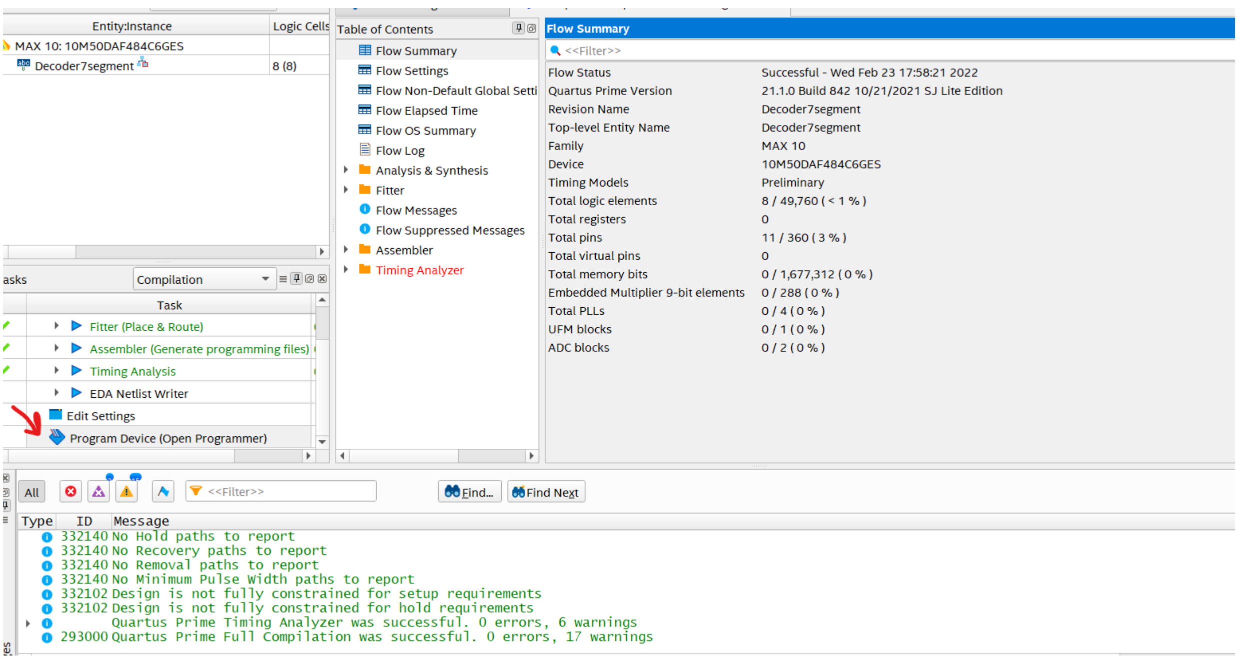
Assign pins on Quartus, open the assignment tab, click on pin planner, and assign pins as figure below.



Node Name	Direction	Location	I/O Bank	VREF Group	I/O Standard	Reserved	Current Strength	Slew Rate	Differential Pair	Strict Preservatio
A	Input	PIN_c10	7	B7_N0	2.5 V (default)		12mA (default)			
B	Input	PIN_c11	7	B7_N0	2.5 V (default)		12mA (default)			
C	Input	Pin_D12	7	B7_N0	2.5 V (default)		12mA (default)			
D	Input	Pin_C12	7	B7_N0	2.5 V (default)		12mA (default)			
led_a	Output	Pin_C14	7	B7_N0	2.5 V (default)		12mA (default)	2 (default)		
led_b	Output	Pin_E15	7	B7_N0	2.5 V (default)		12mA (default)	2 (default)		
led_c	Output	Pin_C15	7	B7_N0	2.5 V (default)		12mA (default)	2 (default)		
led_d	Output	Pin_C16	7	B7_N0	2.5 V (default)		12mA (default)	2 (default)		
led_e	Output	Pin_E16	7	B7_N0	2.5 V (default)		12mA (default)	2 (default)		
led_f	Output	Pin_D17	7	B7_N0	2.5 V (default)		12mA (default)	2 (default)		
led_g	Output	Pin_C17	7	B7_N0	2.5 V (default)		12mA (default)	2 (default)		
<<new node>>										

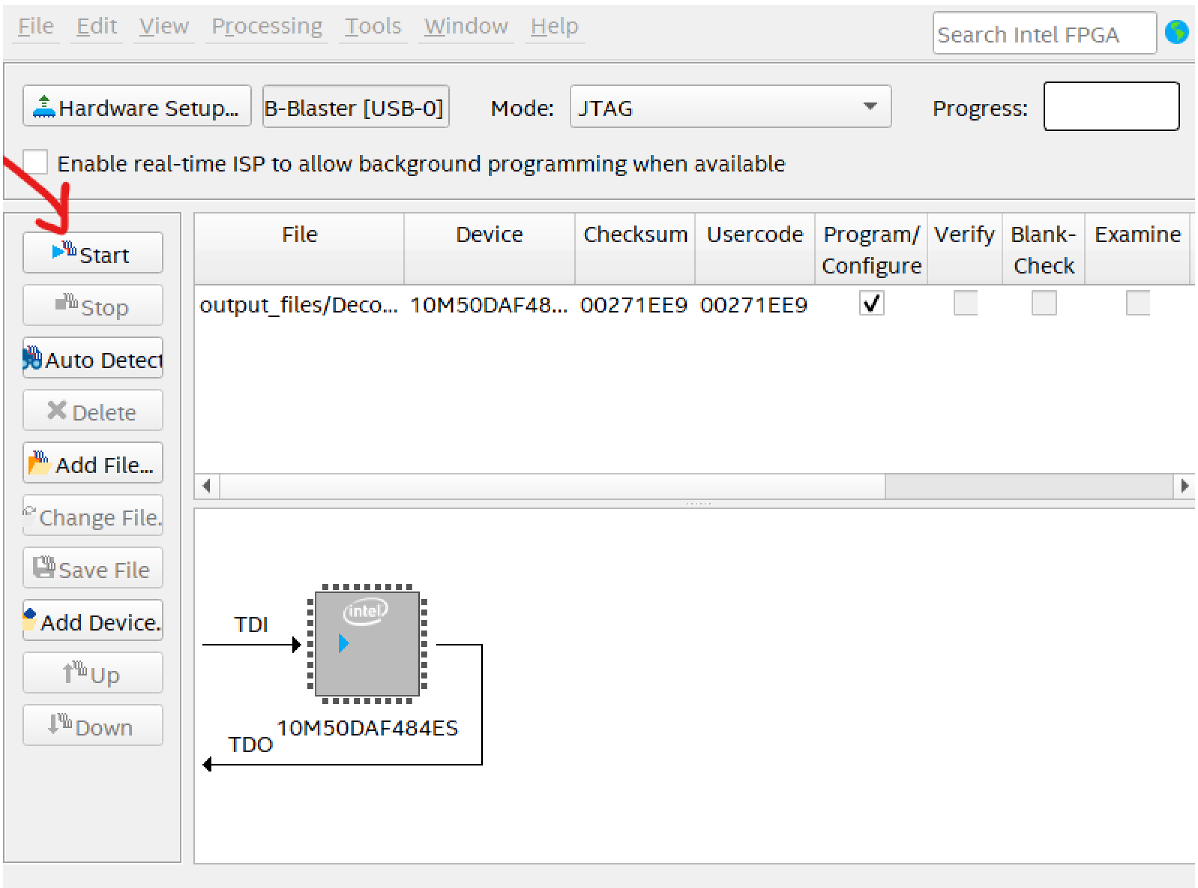
<u>Step 11:</u> Compile all project after pin assignment, like step 9. Program the FPGA.
<u>Step 12:</u> load the program to the FPGA.

Press “Start”

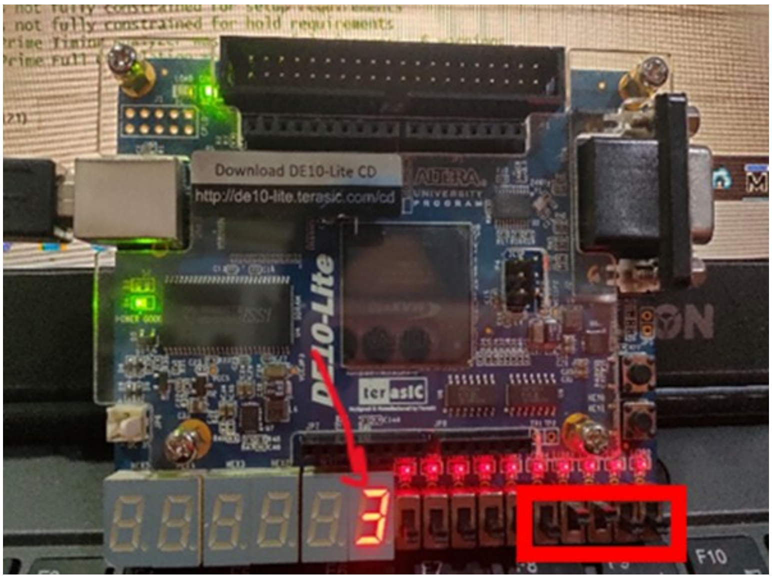


Hint: you may have a problem with the FPGA driver. Check the "device manager" and update the USB driver.

Step 13: Verify the decoder by changing the inputs (switches)



and check the outputs (seven-segments display).


not fully constrained for hold requirements
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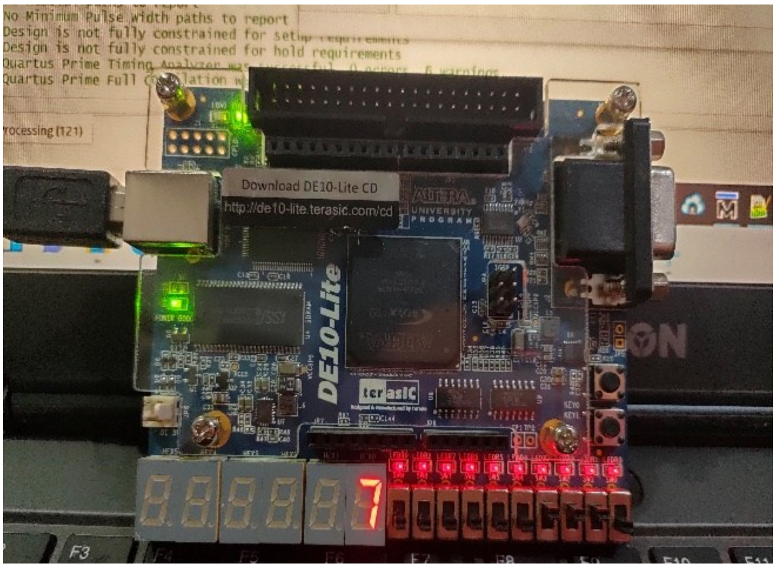
No Minimum Pulse Width paths to report
Design is not fully constrained for setup requirements
Design is not fully constrained for hold requirements
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